AprilTag: Robust Fiducial Markers Detection and Following

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***Abstract*—Navigation is one of the most challenging competencies required of a mobile robot especially in the industrial applications.For efficient navigation, robots must properly estimate their position (localization), detecting any information regarding environmental changes (cognition) and safely drive their motors to reach a target position (motion control) In this paper, a complete apriltag- tracking system will be implemented using Petoi-Dog robot , by detecting a dynamically moving apriltags, estimating the robot position and controlling the robot to track those apriltags.**

***Index Terms*—Petoi Dog, April Tags, Localization, Moving Average Filter, Servo control**

I. Introduction

Path following robots have many real-world applications. They are often used to conduct undertakings in hazardous environments (e.g. Mars's surface, Closed construction sites, underwater). "... there are numerous signs that conventional construction technology is reaching its limits (constant productivity, increasing rate of construction defects, inability to cope with highly complex buildings,extremely low RD spending, etc.)" [1]. Therefore, an increased integration of automated robots in the field of construction is required.Navigation and path following requires both localization and control to be realized. Localization is the process of computing the position of an object with respect to a coordinate frame. Such algorithms use available information about an object's position and state to determine the its position relative to a specified coordinate frame. For this project, AprilTags are used to localize a Petoi Dog, which is an open source programmable robot dog Bittle. AprilTags are unique two-dimensional markers, that can reliably be detected by camera [2]. The hardware consists of a Raspberry Pi 3A+ and a NyBoard V1, along with a camera, Lidar and IMU sensor(see Fig.1).

Often,localization measurements from such tags are noisy and must be filtered. For this, a moving average filter is used. Assuming our motion isn't excessive between frames, such a filter would help in smoothing the noise from our readings. To then control the motion of our robot, the leg servo motors angles would be initiated to reach the target goals.

II. Methodology

A. Camera Calibration

A camera basically converts the light particles which hit the camera's lens and image sensor to digital images which consist of pixels, which are the smallest units of digital images.

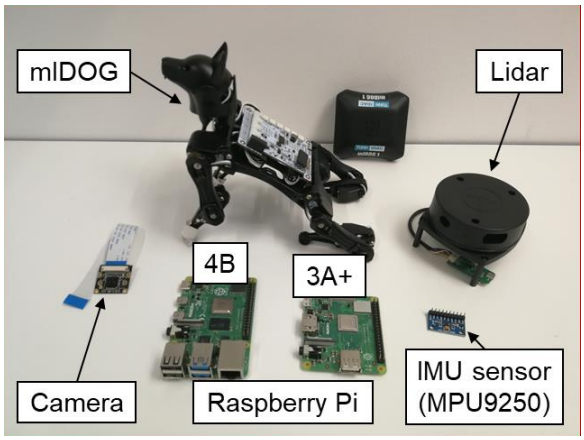


Fig. 1. MlDog Hardware components

There are various camera options used in robotics. Some examples are RGB cameras and 3D cameras, nevertheless, none of those cameras are perfect to transform the real world into digital images [3].

Camera parameters include intrinsic, extrinsic, and distortion coefficients. To estimate the camera parameters, one need to have 3-D world points and their corresponding 2-D image points.

Explaining camera calibration requires a camera model which explains the relationship between a point in the world and the projection on the image plane . The most basic one is the camera pin hole model.

1) Pinhole Camera Model: A pinhole camera is a simple camera without a lens and with a single small aperture. Light rays pass through the aperture and project an inverted image on the opposite side of the camera. Think of the virtual image plane as being in front of the camera and containing the upright image of the scene. [10]

The pinhole camera parameters are represented in a 3-by-4 matrix called the camera matrix. This matrix maps the 3-D world scene into the image plane.

Those parameters are , the focal length which is the distance between the pinhole and the image plane, camera center which represents The coordinates of the center of the pinhole, beside

the other parameters, which will be discussed in more details later.

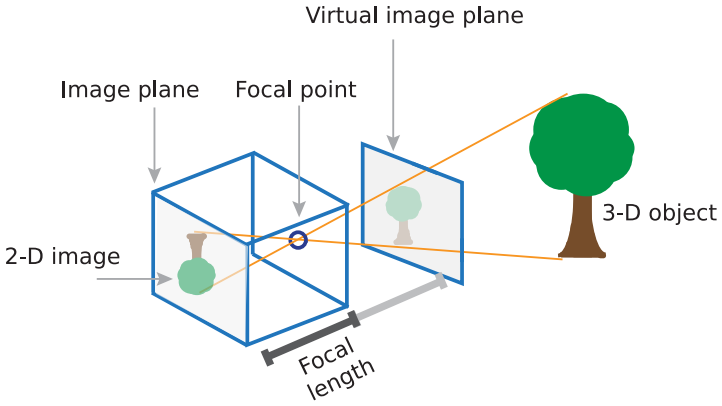


Fig. 2. Pin hole camera model, [10]

2) Coordinate Transformation : The calibration algorithm calculates the camera matrix using the extrinsic and intrinsic parameters. The extrinsic parameters represent the location of the camera in the 3-D scene. The intrinsic parameters represent the optical center and focal length of the camera.

So, in order to project the point in the world frame to the camera image plane, one needs to estimate a transformation matrix that transforms the world frame into the camera frame, which is the fundamental idea of camera calibration that one is given a known set of points in the world and their corresponding projections in the image, and the main task is to find the camera matrix that is responsible for the projection transformation.

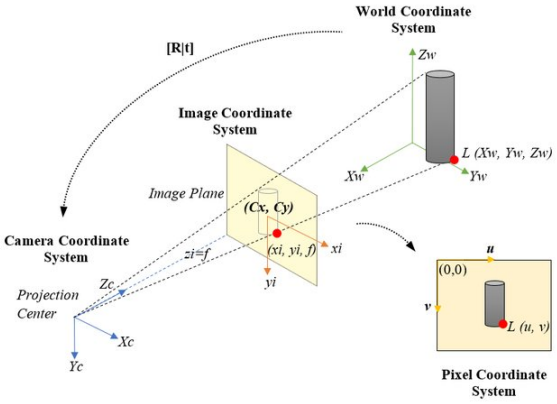


Fig. 3. Simplified Camera model Projection [11]

3) Extrinsic Parameters: The extrinsic parameters consist of a rotation, R, and a translation, t. The origin of the camera's coordinate system is at its optical center and its x- and y-axis define the image plane.

In other words, transformation from the world coordinate system to the camera coordinate system is called camera extrinsics or camera pose.

So, the camera extrinsic matrix M can be represented in the given form:

$$M = \begin{bmatrix} R_3x3 & t_3x1 \\ 0_1x3 & 1_1x1 \end{bmatrix} \quad (1)$$

4) Intrinsic Parameters: The intrinsic parameters of a camera depend on how it captures the images. Parameters such as focal length, aperture, field-of-view, resolution, etc govern the intrinsic matrix of a camera model. So, generally, The focal length and the camera center are the camera intrinsic parameters, K.

$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

So, in abstract, extrinsic parameters give the orientation and location of the camera frame with respect to the real-world frame. On the other hand, intrinsic parameters consist of the internal parameters of the camera and the relationship between camera coordinates and coordinates of the pixels of the digital image in the image frame.

The extrinsic parameters are rotation (3x3) and translation (3x1) matrices, whereas the intrinsic parameters are focal length, skew factor, the optical center of the image, etc. The intrinsic parameters can be shown in a Calibration Matrix, K where f is the focal length and c is the center pixel [1].

5) Distortion in Camera Calibration: Distortion, is a noticeable problem and crucial factor that should be considered during camera calibration.

The lens distortion calibration rectifies the coordinates before they are passed to the spatial calibration. The camera matrix does not account for lens distortion because an ideal pinhole camera does not have a lens. To accurately represent a real camera, the camera model includes the radial and tangential lens distortion.

The physical flaws contained by cameras cause distortions in the images. There exist different forms of distortions. Two significant ones are radial distortion, straight lines appear curved, and tangential distortion, where the actual image appears stretched or tilted in the digital image [4]. The visual examples of the distortion types can be seen in Figure 5.

6) Implementation: Chessboard calibration is a popular method for camera calibration due to being a flat surface, consequently not having the z-axis, and having easily detectable corners and points which appear at straight lines using the Hough transform or many other respectable methods such as Kalman filters.

There are different approaches to camera calibration, the most common ones are Zhang's Method, Tsai's method, and Direct Linear Transformation. Zhang's method uses a checkerboard pattern which needed to be located in at least more than two different positions due to the number of unknowns in the equation.

After extracting the corners of the checkerboard pattern, the method calculates a projective transformation between the captured image points and a scaling factor. Later on, the intrinsic and extrinsic parameters are calculated by solving the equations. On the other hand, Tsai's method has an assumption that some camera parameters are given, for instance by the camera provider, which significantly reduces the errors in the initial estimation guesses [6], [7], [8].

In this paper, OpenCV-python library will be used to calibrate the camera. 20 pictures of a 7x9 checkerboard with different positions are collected with miDOG's pi-camera and stored in an external file. Each image is grey scaled and corners of the checkerboards are found using OpenCV's findChessboardCorners function. After, using calibrateCamera function which utilizes Zhang's method, the calibration matrix for intrinsic parameters and the distortion coefficients are calculated.

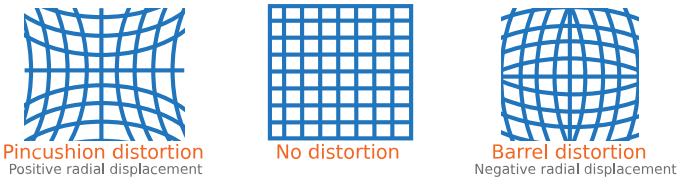


Fig. 4. Different camera distortions [13]

B. Environment setup

Before starting with AprilTag detection process, one need to set-up the environment. This set-up initially includes setting up a global coordinate frame, which in our case , is the bottom left corner of the room with the X-axis pointing to the right, Y-axis upwards and Z-axis out of the wall. Some pairs of AprilTags also placed across the room. Furthermore, measuring the position and the orientation(distance and Euler angles) of the AprilTags with respect to the global frame, which will be needed later on , to obtain the pose of the camera with respect to the global frame.

C. AprilTag Detection

AprilTag is a visual fiducial system, useful for a wide variety of tasks including augmented reality, robotics, and camera calibration. Targets can be created from an ordinary printer, and the AprilTag detection software computes the precise 3D position, orientation, and identity of the tags relative to the camera. [12]

There are various families of AprilTags, with different advantages and disadvantages. In this paper, 36h11 will be used , as it is one of the most common and efficient AprilTags.

In this section, AprilTag detection process will be explained. To detect an AprilTag in an image, AprilTag and OpenCV libraries will be used.

The detection function consists of two main algorithms which are fast corner detection and quadrilateral detection. Fast corner detection algorithms are used to quickly identify

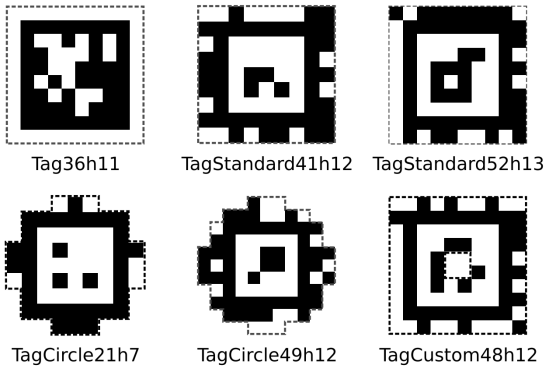


Fig. 5. AprilTags Families [12]

potential corners of AprilTags in the image, later they are used as starting points for quadrilateral detection. Once potential corners have been identified, quadrilateral detection algorithms search for quadrilaterals which are basically four-sided shapes in the image that match the expected shape of AprilTag markers [9].

Once a marker is detected, the marker's binary payload is decoded to determine the marker's identifier, and the position and orientation of the marker are estimated using methods such as perspective-n-point (PnP) algorithms. Finally, the detected marker is validated using various checks such as verifying the parity of the marker's identifier and verifying that the marker's quadrilateral has the correct shape.

Before starting to apply these algorithms, the detection function requires the images to be preprocessed by modifying them to reduce noise and enhance contrast. Which helps the fast corner detection algorithm to operate smoothly.

D. Localization and Filtering

Upon using the detect function, we get the Pose of the tags in the camera frame. Now that one has this information, one can localize the robot with respect to the global frame.

To achieve this, the camera frame should be transformed to the tag frame. By multiplying the rotation matrix of the tag with respect to the camera and the translation vector of the tag with respect to the camera, one can get the exact location of the camera in the tag frame. Finally, based on the measurements and angles from the environment setup of the global frame with respect the tags, one then transforms the tag frame to the global frame and thus localize the robot with respect to the global frame. Since pairs of tags are used, the position of all tags are averaged to get the estimated position.

Since the reading from the AprilTags are quite noisy with lower accuracy when far away from the tags, at 3m approximately, will be filtered using a moving average filter at constant time steps to smoothen these readings and improve the measurements.

E. Control System

The main objective of this paper is to track a moving AprilTag which requires some information about the camera model (K matrix, distortion) and the relative position of the camera with respect to the AprilTag local frame. In this section , the complete control system will be illustrated with its main components.

Fig. 6. AprilTag Following Process

The control process starts by detecting a fiducial marker in front of the camera , then comes the detector job to estimate the relative position of the camera and pass it to the controller. After that, the switching controller will compare the distance between the camera and AprilTag , and an initialized safe distance , and if the distance is less than the loaded safe distance , the controller will temporarily stop the robot until the error value increases again when the camera moves away from the AprilTag.

So, the main controller task is to calculate the current error and send a command signals to the robot servo motors, to decrease that error or in other words to decrease the distance to the camera.

Following AprilTag, requires a path to be generated, and that path might be straight, curved, or a combination of both. While there are many ways to design paths, one popular method is to utilize way-points, which is an ordered list of points that have to be tracked in an ordered sequence.Without any more details, way-points generally includes specification about velocity and acceleration in the case of path tracking that requires following path points at a specific time. In this paper, only path-following will be presented. Hence, the path points only include information about the required 3D position of the AprilTag.

III. Results

IV. Conclusion

This is conclusion

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